IP addressing

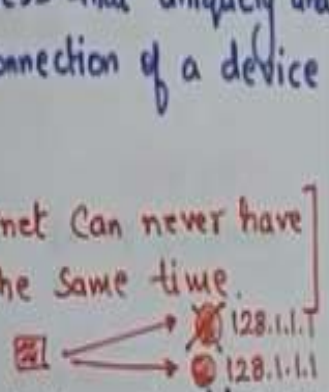
IPV4 and IPV6

Network Layer: Logical Addressing [IPV4 Addresses]

An IPV_4 is a 32-bit address that uniquely and universally defines the connection of a device to the Internet.

[Two device on the Internet can never have the same address at the same time.]

IMP. TERMS:



(i) Address Space: It is the total no. of addresses used by the protocol.

'n' bits in a add.

↳ $n = 32$ ↳ 2^n addresses.

Add. space (IPv4) : 2^{32}

$= 4,294,967,296.$

(ii) Notations:

↳ (a) Binary Notation: In this IPV_4 is displayed as 32 bits or 4 byte.

01110101 10010101 00011101 11101010

↳ (b) Dotted-Decimal Notation: Used to make IPV_4 more compact and easier to read.

10000000 00001011 00000011 00011111

↳ 128. 11. 3. 31

↳ (c) Hexadecimal Notation: Each Hexadecimal digit is equivalent to four bits. This means that 32-bit address has 8 hexa-decimal digits.

Classful Addressing: In this the address space is divided into 5 classes
- Class A
- class B
- class C
- Class D
- class E

IMP.:

How to find Class of an Address:-

Binary Notation	First Byte	IMP.: Dotted-Decimal	First Byte
class A	0....	Class A	0-127
class B	10....	class B	128-191
class C	110....	class C	192-223
class D	1110....	class D	224-239
class E	1111....	class E	240-255

Netid and Hostid: Only class A, B or C is divided into netid and hostid.

class	Byte1	Byte2	Byte3	Byte4
classA	Netid	Hostid	Hostid	Hostid
ClassB	Netid	Netid	Hostid	Hostid
Class C	Netid	Netid	Netid	Hostid

n bits = 1

(32-n) = 0

A: n = 8

B: n = 16

C: n = 24

Mask: It helps to find netid and hostid.

Class	Binary (Mask)	Decimal
classA	11111111 00.....	255.0.0.0
class B	11111111 11111111 00....	255.255.0.0
classC	11111111 11111111 11111111 0...	255.255.255.0

DEFAULT MASK FOR CLASSFUL ADDRESSING

(Decimal Dotted)

Ques1.) Find the Class of each Address:-

Byte1

a.) 227.12.14.87 ⇒ Class D

b.) 193.14.56.22 ⇒ class C

c.) 14.23.120.8 ⇒ class A

d.) 252.5.15.111 ⇒ Class E

Ques2.) Find the class of each Address:-

Binary Notation.

Byte1

a.) 00000001 00..... → class A

b.) 11000001 100..... → class C

c.) 10100111 110..... → class B

d.) 11110011 100..... → class E

0

10-

110-

1110-

1111-

Classless Addressing:- In this, Variable length blocks are used that belongs to no class.

Restrictions:-

↳ (i) Addresses in a Block must be Contiguous.

(ii) No. of Addresses in a Block must be a power of 2 [1, 2, 4, 8, 16, ...]

(iii) The first address must be evenly divisible by the no. of addresses.

Block

First → 205.16.37.32 → $16 = 2^4 =$

↳ 205.16.37.33

↳ 205.16.37.34

⋮

Last → 205.16.37.47

Mask in classless Addressing Can take any Value from 0 to 32. (n) = 1, (32-n) = 0.

↳ Slash/CIDR Notation.

x.y.z.t/n → defines the mask.

IMP Points:

(i) First Address in block can be found by setting the rightmost (32-n) bits to 0s.

(ii) Last address can be found by setting the rightmost (32-n) bits to 1s.

(iii) No. of Addresses: 2^{32-n}

Example:- One of the Addresses is 205.16.37.39/28.

Find first, last and total no. of addresses.

11001101 00010000 00100101 00100111 → 00101111

F.A:- ↓ ↓ ↓ 00100000 N.A = 2^4

205.16.37.32 L.A = 205.16.37.47 = 16

One More Way of Extracting Block Info^n:-

↳ (i) No. of Addresses, $N = 2^{32-n}$.

(ii) First Address = (Address) AND (MASK)

(iii) Last Address = (Address) OR (COMPLEMENT of MASK)

Example:- For a Address 205.16.37.39/28, Find

i) First Address

ii) Last Address

iii) No. of Addresses. $= 2^{32-28}$

$= 2^4$

$= 16$

Address:- 11001101 00010000 00100101 00100111

Mask:- 11111111 11111111 11111111 11110000

First Add. 11001101 00010000 00100101 00100000

AND

1+1=1

Mask Comp. 00000000 00000000 00000000 00001111

11001101 00010000 00100101 00101111

IPv4 Protocol: IP Packet/Datagram Format

IMP Points:-

(i) Used by TCP/IP protocols at N/W Layer.

(ii) Unreliable and Connectionless

Header	Data

Header Length

(Header + data)

(IPv4)

VER (4)	HLEN (4)	Service (8)	Total Length (16)
Identification (16)		Flags (3)	Frag. offset (13)
Time to live (8)		Protocol (8)	Header Checksum (16)
Source IP address (32)			
Destination IP address (32)			
option (32)			

Max. time for which IP data gram is alive

Error

Higher Level

TCP UDP ICMP...

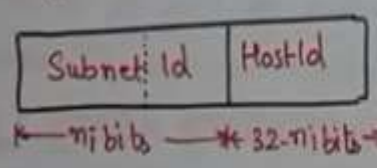
(Header) [32 bits]

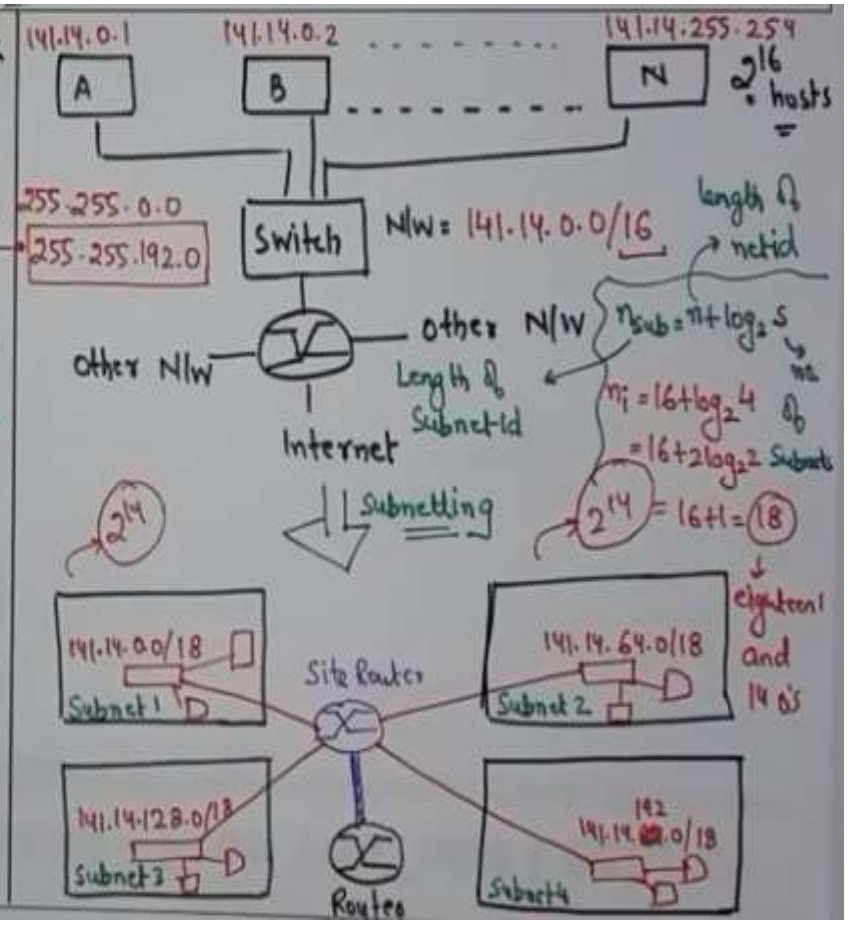
Subnetting:- It is the idea of splitting a block to smaller blocks, In this a N/W is divided into several smaller subnets (SubN/Ws) with each subnetwork having its own subnetwork address.

↳ A SubN/W has subnetid and Hostid.

[Subnetting ↑ the length of netid and ↓ length of hostid or ↑ no. of 1's in the mask.]

Subnet Address:- When a N/W is subnetted, the first address in the subnet is identifier of the subnet and is used by the router to route the packets destined for that SubN/W.





Super Netting:- In this an organization can combine several class 'C' blocks to create a larger range of addresses. In other words "Several N/Ws are combined to create a Super N/W."

[Supernetting ↓ the no. of 1s in the Mask]

Super Net Mask:- It is a reverse of subnet mask.

Length of Super Net Id

$$n_{super} = n - \log_2 C$$

n_{super} → Length of Supernet Id

n → n = length of Net Id

C → no. of Class C Blocks that are combined.

Default Mask:- $n = 24$

11111111 11111111 11111111 00000000

Let' say we are breaking into 8 subnets.

$$n_i = n + \log_2 S$$

$$n_i = 24 + \log_2 8 \rightarrow \log_2 2^3 = 3\log_2 2 \rightarrow 1$$

$n_i = 27$ one's in Subnet Mask

11111111 11111111 11111111 11100000] Subnet mask

Combining 8 class C to form super Net

$$n_s = n - \log_2 C \Rightarrow 24 - \log_2 8 = 21$$ one's in Supernet Mask

11111111 11111111 11111000 00000000] Subnet mask

IPv6 Addresses:- It is of 128 bits or 16 bytes.

↳ Length is (4) times the length address of IPv4.

Notations:-

(i) **Dotted-Decimal:-** It is used for IPv4 compatibility. 221.14.65.11.105.45.170.34.12.234.18.0.14.0.115.255](16)

MP (ii) **Colon Hexadecimal:-** It is used to make the address more readable. In this notation, the 128 bits are divided into 8 sections, each of 2 bytes in length. [Two bytes in Hexadecimal req. 4 Hexadecimal digits].

FDEC: BA98: 7654: 3210: ADBF: BBFF: 2922: FFFF

Abbreviation:- It is a technique to reduce the length of IPv6 address. It is done by omitting/removing the leading zeros of a section.

[NOTE:- Only the leading zeros can be dropped]

[Zero-Compression]

FDEC: 0074: 0000: 0000: 0000: B0FF: 0000: FFF0

omitting these zeros

FDEC: 74: 0: 0: 0: B0FF: 0: FFF0 } Abbreviated Address

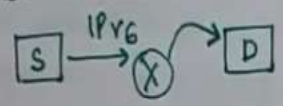
↓

FDEC: 74: : B0FF: 0: FFF0

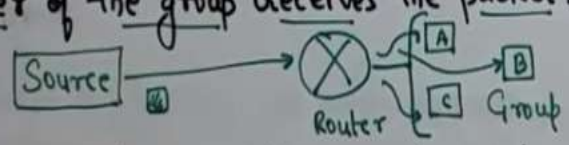
↳ GAP.

Types of Address Space in IPv6:-

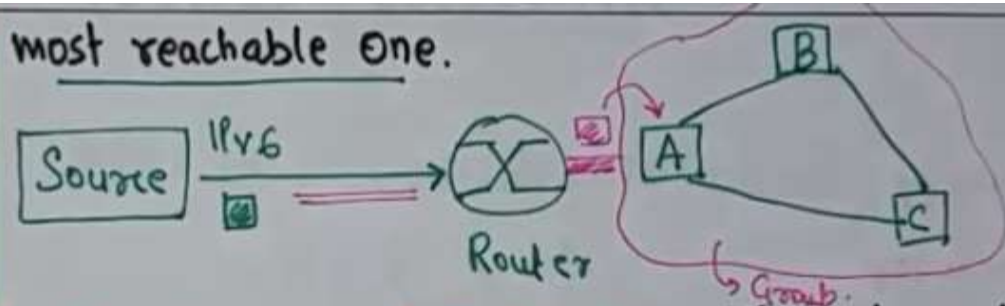
(i) Unicast Addresses:- It defines single interface or Computer. The packet Sent to a unicast address will be routed to the intended PC or recipient.



(ii) Multicast Addresses:- These are used to define a group of computers/hosts. In this, each member of the group receives the packet.



(iii) Anycast Addresses:- Defines group of nodes or Computers that all share a single address. A packet with anycast address is delivered to only one member of the group which is the most reachable one.

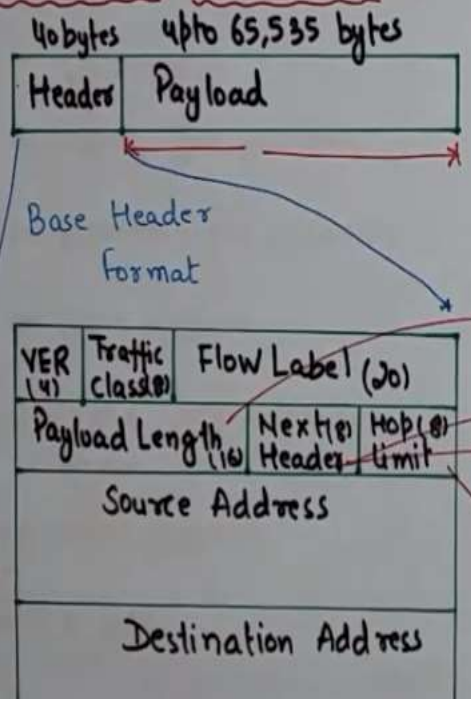


(iv) Broadcasting and Multicasting:- IPv6 doesnot define broadcasting and considered it as a special case of multicasting.

Reserved Addresses:-

↳ Starts with 8 0's

IPv6 Protocol:- Packet Format



VER:- IPv6 = 6

Traffic class:- distinguishing diff. Payload

Flow Label:- provide special handling for particular data flow.

Length of IP datagram – Base header

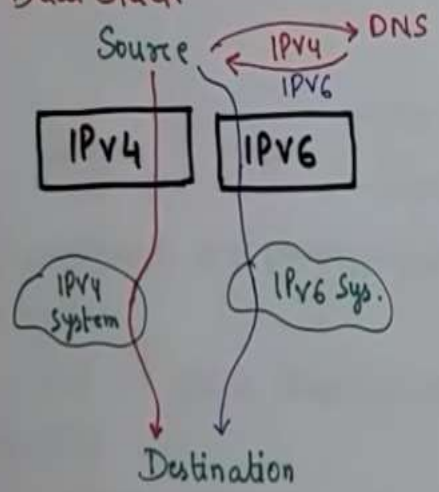
2 = ICMP

6 = TCP, 17 = UDP ....

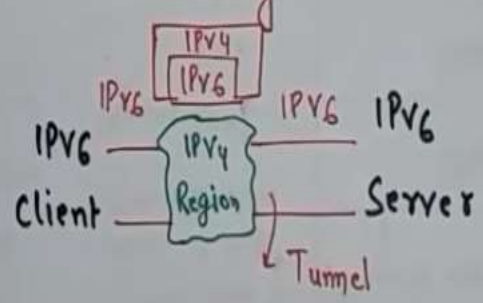
TTL

Transition from IPv4 to IPv6:- There are three transition strategies.

Dual Stack



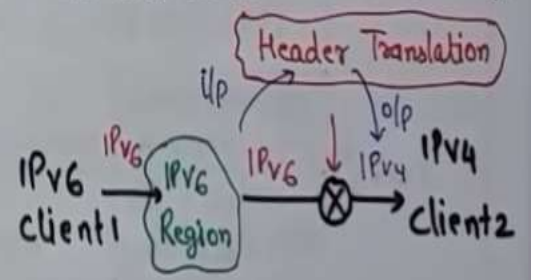
Tunneling



IPv6 Packet is encapsulated in an IPv4 Packet when it enters the IPv4 Region.

Header Translation

When most of system are on IPv6 but some still uses IPv4.



DIFFERENCE BETWEEN IPv4 AND IPv6 IMP..

IPv4	IPv6
1.) Length of Address: 32 Bit	i) Length of Address: 128 Bit
2.) Represent in Decimal notation	ii) Represented in Hexadecimal notation
3.) IPsec Support: Optional	3.) IPsec Support: Inbuilt
4.) Packet Flow Indication: NONE	4.) Packet Flow: YES → FLOW Label Field
5.) checksum field: YES	5.) checksum field: NONE
6) option field: YES	6) options field: NONE, IPv6 Extension Header ✓
7.) Address (IP) to MAC = (ARP)	7.) Replaced By: Neighbour Discovery (NDP) Protocol
8) Broadcast Message: YES	8) Broadcast Message: Special type of Multicast Addr.
9.) Total No. of Addresses: 2^{32}	9.) Total No. of Addresses: 2^{128}

IP: 192.168.10.0

Subnet mask: 255.255.255.224

11111111.11111111.11111111.11100000

Find out

1. Block size ?
2. Number of subnets ?
3. Number of valid hosts ?
4. what are the valid subnets or subnets ID ?
5. what are the first valid host & Last valid host?
6. what's the Broadcast address ?

1. Block size = 256 − 224 = 32
2. Number of subnets = 2^x ← Number of 1's
   = 2^3 = 8 subnet
3. valid host = $2^y - 2$ ← number of 0s
   = $2^5 - 2$
   = 32 − 2
   = 30

192.168.10.0

subnet ID	0	32	64	96	128	160	192	224
First V.H.	1	33	65	97	129	161	193	225
L.V.H.	30	62	94	126	158	190	222	254
Broadcast	31	63	95	127	159	191	223	255

⊛ Suppose you are given an IP address of 192.168.10.0, Subnet mask 255.255.255.248, write down all subnet address, First valid host, Last Valid host and broadcast address in each subnet.

Soln

IP address – 192.168.10.0
Subnet mask – 255.255.255.248

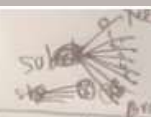
Binary representation – 11111111. 11111111. 11111111. 11111000

⊛ subnet ID
⊛ First valid host
⊛ Last valid host
⊛ broadcast address

① Block size = 256 − 248
= 8

② number of subnet = 2^x ← number of 1's
= 2^5
= 32

③ number of valid host = $2^y - 2$ ← number of 0's
= $2^3 - 2$
= 8 − 2
= 6



192.168.10.0
255.255.255.248 (32)

Subnet ID	0	8	16	24	32	40	48	56
First valid host	1	9	17	25	33	41	49	57
Last valid host	6	14	22	30	38	46	54	
Broad-cast address	7	15	23	31	39	47	55	

⑧ 175.231.232.116/27 ← CIDR classless Inter Domain Routing

11111111.11111111.11111111.11100000

① Subnet → 255.255.255.224

② Block size = 256 − 224
= 32

③ Number of subnets = 2^x ← number of 1s
= 2^3
= 8

① what is the subnet mask?
② what is the Block size?
③ Number of subnets?
④ Number of hosts?
⑤ what are the subnet addresses?
⑥ First valid host & Last valid host
⑦ Broadcast address

④ Number of hosts = $2^y - 2$ ← 0's
= $2^5 - 2$
= 30

175.231.232.116

Subnet ID	0	32	64	96	128	160	192	224
First V.H.	1	33	65	97	129	161	193	225
Last valid host	30	62	94	126	158	190	222	254
Broadcast address	31	63	95	127	159	191	223	255

(8) 22.204.134.249/20

(1) Subnet mask?

(2) Block size?

(3) Number of subnets?

(4) Number of hosts (valid)?

(5) what are the subnet ID's?

(6) First & last valid host?

(7) what's the Broadcast address?

22.204.134.249

11111111.11111111.11110000.00000000

(1) subnet mask: 255 . 255 . 240 . 0

(2) Block size = 256 − 240 = 16

(3) Number of subnets = 2^{x} ← 1's

= 2^{4} = 16 subnets

(4) Number of valid hosts = 2^{y} − 2 ← 0's

= 2^{4} − 2

= 14

Net ID − 22.204.128.0

F.V.H → 22.204.128.1

L.V.H → 22.204.143.254

B.A → 22.204.143.255

Subnet address	0.0	16.0	32.0	48.0	64.0	80.0	96.0	112.0	128.0	144.0	160.0	176.0	192.0	208.0	224.0	240.0	
First valid host	0.1	16.1	32.1	48.1	64.1	80.1	96.1	112.1	128.1	144.1							
Last valid Host	15.254	31.254	47.254					127.254	143.254								
Broadcast address	15.255	31.255	47.255					127.255	143.255								